Latent Render-and-Compare: SE(3) Diffusion with a Feature-Field World Model

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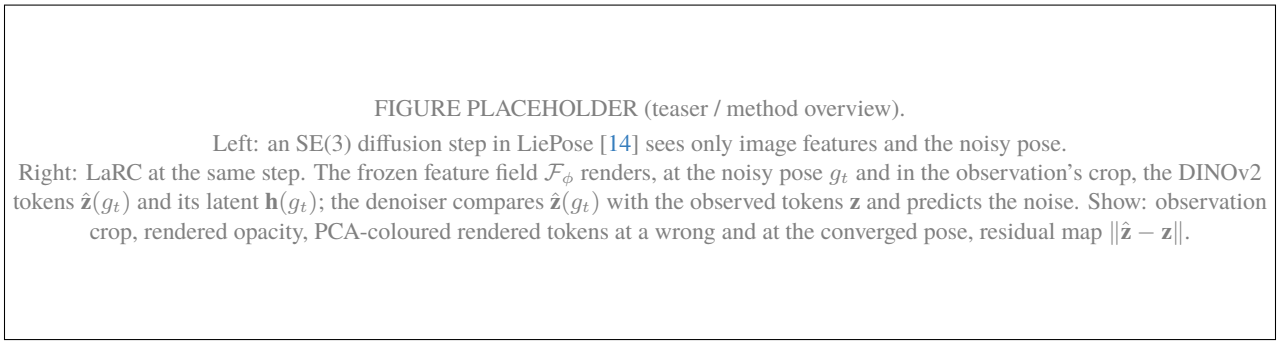


Figure 1. **Render-and-compare inside every denoising step, in feature space.** At each step of SE(3) diffusion, a frozen feature field renders what the object would look like, as DINOv2 tokens, at the current noisy pose hypothesis; the denoiser conditions on the comparison between this rendering and the observation. No reference image, textured model or test-time optimisation is needed.

Abstract

Diffusion models on SE(3) handle the pose ambiguity of symmetric and occluded objects by sampling. Each denoising step, however, is a feed-forward function of image features and the noisy pose, so the network never checks whether its current hypothesis explains the image. Render-and-compare refiners do perform this check. They need a renderer and a textured 3D model, though, and run as a separate stage after an initial estimate. We propose LaRC, a pose-estimation architecture that carries out this comparison in a learned feature space inside every denoising step. Its world model is a feature field of the object: tri-planes and a small MLP, fitted once to the frozen DINOv2 tokens of posed training crops. It renders by volume rendering the tokens that the object would produce at any pose hypothesis, in the observation's crop. The field needs no reference image, CAD mesh or RGB reconstruction, and its 3D structure is learned only from the tokens. At every denoising step and for every sample, the denoiser receives the observed tokens, the tokens rendered at its current hypothesis, their difference, and the field's latent at the surface each ray hits. A control that removes only the rendered quantities isolates the effect of the comparison. We evaluate on T-LESS under the BOP protocol, on the SPEED+ spacecraft benchmark with its official score, and against LiePose trained with the same budget. A retrieval-referenced world model, which predicts tokens from the nearest training view, is studied in the supplement. ***[TODO: one sentence of results, written only after the tables are filled.]***

1. Introduction

Estimating the 6D pose of a known object from a single RGB image is ambiguous when the visible part of the object looks the same from several viewpoints. Symmetry causes this, as do occluded distinguishing features and texture-less surfaces that offer little to disambiguate [10, 12, 29]. Treating the pose as a distribution rather than a single value handles the ambiguity, and diffusion models on SE(3) have become an effective way to represent such distributions. LiePose [14] samples poses with a score-based diffusion process on SE(3) and reaches high accuracy on T-LESS without symmetry annotations.

A diffusion denoiser, however, is an open-loop function. At each step it maps image features and the current noisy pose to a correction, and it never checks whether the current hypothesis is consistent with the image. That check is the core of render-and-compare refinement [19, 20, 23, 46]: ren-

der the object at the hypothesis, compare the rendering with the observation, and update. Render-and-compare has been the most accurate final stage in BOP for years [13, 42]. It requires a renderer and a textured 3D model, however, runs as a separate refiner after an initial estimate, and does not by itself represent a multi-modal distribution.

We ask whether the comparison can be moved into the diffusion process, in a learned feature space, so that every denoising step of every sample is conditioned on how well its hypothesis explains the image. This needs a model that answers "what would the image features be if the object were at pose g?" for arbitrary g, cheaply enough to be queried at every step. We build such a model as a world model in the sense of predictive latent models [3, 4, 50], with the object pose as its action. It operates on frozen DINOv2 [32] patch tokens, and it is a *feature field*: tri-plane features [5] and a small MLP give, at every point in a box around the object, a density and a latent vector. The field renders the tokens at a pose by volume rendering [28] rays through the observation's crop, and it is fitted once per dataset to the tokens of posed training crops. Everything it renders is a DINOv2 token or a latent. It has no RGB output and is given no surface, mesh or texture; its geometry is whatever density best explains the tokens. Because the field renders a pose rather than borrowing a nearby training view, a small rotation of the hypothesis moves every ray by exactly that rotation. Its prediction carries no lighting, background or occlusion of another image.

Our method, LaRC (Latent Render-and-Compare), is a diffusion-based pose estimator built around this world model (Fig. 1). It replaces the usual image-encoder-plus-pose-MLP denoiser with a closed loop. At every step the field renders, for each sample's current pose, the tokens in the observation's crop and the field's latent at the surface each ray hits. A token transformer reads the observed tokens, the rendered tokens, their difference and the latent, conditioned on the noise level and the pose. The image encoder is a frozen foundation model, and the object's appearance as a function of pose is held by the frozen field rather than learned into the denoiser. The same rendering defines an energy, the foreground-weighted distance to the observation, which ranks a set of samples at test time without ground truth. We compare against a control that is identical to LaRC except that it receives no rendered quantity, so the contribution of the comparison is measured in isolation.

We evaluate on T-LESS [10] under the BOP protocol [11] and on the SPEED+ spacecraft benchmark [35] with the official SPEC2021 score, against LiePose and the control trained by us with the same data and crops. We also report what the field can discriminate on its own, and compare it with a retrieval-referenced world model that predicts tokens from the nearest training view (Sec. I). **[TODO: summarise the main findings in one paragraph once Tabs. 1–2 are filled; no claim before the numbers exist.]**

Our contributions are:

- A feature-field world model for object pose. It renders frozen DINOv2 tokens at an arbitrary pose by volume rendering a tri-plane field, is fitted once from posed training crops, and needs no reference view or textured model (Sec. 4.2).
- LaRC, a closed-loop denoising architecture for SE(3) diffusion. A frozen foundation-model encoder and a token transformer compare the field's rendering at the current hypothesis with the observation at every step. The same rendering gives an energy for selecting among samples without ground truth (Secs. 4.4 and 4.5).
- A controlled evaluation against LiePose at a matched training budget and a no-world-model control, with ablations of what the denoiser reads from the field and an analysis of what the field can and cannot discriminate (Sec. 5).

2. Related Work

Instance-level 6D pose estimation. Methods for known objects either regress the pose directly [26, 45, 47], or predict 2D–3D correspondences and solve PnP [7, 9, 12, 24, 40], or compare the image with templates or codebooks of rendered views [30, 31, 41]. Symmetric objects make the image-to-pose map one-to-many; most methods handle this with symmetry-aware losses that need symmetry annotations [26, 45]. Like template methods, our world model compares the image with predicted appearance. It renders that appearance at a continuous pose from a learned 3D field, instead of matching against a fixed set of rendered or real views.

Render-and-compare refinement. DeepIM [23] iteratively predicts a pose update from an observed and a rendered image; CosyPose [19], MegaPose [20] and FoundationPose [46] scale this to many objects and to novel objects. RePOSE [15] renders learned deep textures instead of RGB, and PixLoc [38] aligns learned feature maps for camera localisation. These methods compare in image or feature space, but they all need a renderer and a 3D model at test time and act as a refinement stage on a given initial pose. LaRC needs neither. Its "rendering" is the frozen DINOv2 tokens produced by a feature field fitted to posed training crops, and the comparison happens inside every step of a generative sampler, for every sample.

Diffusion and distributions on rotations and poses. Distributions over rotations have been modelled with implicit densities [29], normalising flows and mixtures; diffusion models on SO(3) [21] and SE(3) [44, 49] were introduced for protein structure, grasping and alignment. LiePose [14] applies score-based diffusion on SE(3) to object pose from

images, with a surrogate Stein score and a ResNet image encoder. LaRC uses a different denoiser: a frozen encoder and a world model, so that in addition to observation features the network sees the rendered appearance at its own current hypothesis. Classifier guidance [8] steers a sampler with the gradient of an external model; our world model enters as an input to the denoiser instead, and is used as an energy only to rank finished samples.

Neural fields and pose. Neural radiance fields [28] represent a scene as a density and a colour field rendered by volume rendering; tri-planes [5] make the field compact and fast to query, and per-image appearance codes absorb lighting changes [27]. Distilled feature fields [17] render the features of a pretrained 2D encoder instead of colour. iNeRF [48] estimates a pose by gradient descent on the photometric error of a radiance field, which is local and needs a good initial pose. Our field renders frozen DINOv2 tokens of one object. It is fitted from the training crops of a pose dataset, not from a multi-view capture, and it is not optimised against at test time: the denoiser reads its rendering as an input at every step.

World models on pretrained features. Joint-embedding predictive architectures learn by predicting representations rather than pixels [3, 4]. DINO-WM [50] trains an action-conditioned predictor on frozen DINOv2 patch features and plans by optimising actions so that predicted features match a goal. We use the same principle with the object pose as the "action". The world model maps a pose to the tokens the object would produce there, and the denoiser uses the prediction to reason about its hypothesis.

Spacecraft pose estimation. SPEED [16] and SPEED+ [35] benchmark the pose of a known spacecraft, with SPEED+ testing on hardware-in-the-loop images (lightbox, sunlamp) after training on synthetic renders only. Leading methods regress keypoint heatmaps and solve PnP [6], often with domain randomisation, style augmentation or adaptation on the unlabeled test domains [33, 35, 36]. The spacecraft has near-symmetric configurations that produce flipped estimates, which makes it a natural test of a sampler that can represent and resolve ambiguity.

3. Background: The Generative Process

Pose parametrisation. Following LiePose [14] and GDR-Net [45], a pose is expressed relative to an image crop I of the object: the rotation is allocentric [18], *i.e.* it removes the rotation induced by the object's position in the image, and the translation is scale-invariant [24], $t = (\delta_x/w,\ \delta_y/h,\ z/r)$, with (δ_x, δ_y) the offset of the projected object centre from the crop centre, (w, h) the box size and r the crop's zoom ratio. We write $g = (R, t) \in \mathrm{SE}(3)$ for this crop-relative pose; it maps to the camera frame given only the crop box.

FIGURE PLACEHOLDER (architecture).
(a) World model: pose g + crop camera $\rightarrow$ 16×16 rays through the crop $\rightarrow$ 32 samples per ray in the object box $\rightarrow$ tri-planes + MLP $\rightarrow$ density, latent $\mathbf{h}(x)$ $\rightarrow$ volume rendering $\rightarrow$ head (appearance code of the observation) $\rightarrow$ tokens $\hat{\mathbf{z}}(g)$, opacity, ray latent; 2×2 pooling to 8×8.
(b) Denoiser step: $g_t \rightarrow \mathcal{F}_\phi$ (frozen) $\rightarrow [\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}, w, \mathbf{h}]$ tokens + conditioning token $c(t, g_t) \rightarrow$ 4 adaLN blocks $\rightarrow \epsilon_\theta$.
(c) Selection: N samples, energy $E(g_i)$, arg min.

Figure 2. **Architecture of LaRC.** (a) The world model renders tokens at pose g from a feature field of the object. (b) Each denoising step compares this rendering at the current hypothesis with the observation. (c) The same rendering gives an energy that ranks finished samples.

Diffusion on SE(3). The forward process [14] perturbs a clean pose g_0 by Gaussian noise in the Lie algebra, $g_t = g_0 \operatorname{Exp}(\sigma_t \epsilon)$, $\epsilon \sim \mathcal{N}(0, I_6)$, with a variance-exploding power-law schedule σ_t over $T = 100$ steps. A network $\epsilon_\theta(g_t, t, I)$ is trained to predict the noise, and sampling starts from a uniform pose and applies T DDIM-style [39] updates on the group. Each training image is perturbed into many noisy poses ("slices") per step. In LiePose, ϵ_θ is a ResNet-34 image encoder followed by Fourier-feature MLP blocks that take the noisy pose and the time step.

What the denoiser does not see. ϵ_θ receives the image once, as a feature vector, and the hypothesis g_t as a vector. Whether the object would look like I at g_t is never computed explicitly; the network must learn it implicitly for every pose in SE(3). Section 4 introduces a denoiser architecture that computes it: an explicit prediction of the appearance at g_t, compared with the observation.

4. Method

LaRC has two parts. A world model $\mathcal{F}_\phi$ renders the DINOv2 tokens of an object at an arbitrary pose (Secs. 4.1 and 4.2). It is a feature field, fitted first and then frozen (Sec. 4.3). A diffusion denoiser reads the field's rendering at its current hypothesis at every step (Sec. 4.4), and the same rendering ranks complete samples (Sec. 4.5). Figure 2 gives an overview.

4.1. Observation tokens

The input is a 224×224 crop I of the object, cut and resized as in LiePose (dynamic zoom-in at training time). Pixels outside the visible mask are set to zero when a mask is part of the protocol (Sec. 4.6). A frozen DINOv2 ViT-S/14 [32] maps the crop to 16×16 patch tokens of dimension 384, which we average-pool 2×2 into $\mathbf{z} \in \mathbb{R}^{64\times384}$. A per-token weight $w_p = w_0 + m_p$ emphasises the object: $m_p \in \{0, 1\}$ is the max-pooled visible mask dilated by one token, and $w_0 = 0.05$ keeps a small weight on background tokens. Where no mask is available (SPEED+), $m_p = 1$. The crop is described by its *crop camera* k: the zoom, the crop origin, the box size and the image's intrinsics. k contains no pose information, and it maps a crop-relative pose (Sec. 3) to the camera frame. Tokens of all training crops are computed once and cached.

4.2. A feature-field world model

The world model must say what the tokens of the object would be at any pose, in the observation's crop. We represent the object by a field over a box of half-size r_o around its origin. Tri-plane features [5] (three 128×128 planes with 32 channels, bilinearly sampled and summed) and Fourier features of the point position enter a two-layer MLP of width 128, which outputs a density $\sigma(x) \geq 0$ and a latent $\mathbf{h}(x) \in \mathbb{R}^{128}$; outside the box the density is zero. On T-LESS each object has its own planes, and the MLP and the heads below are shared by all objects.

Rendering. Given a pose hypothesis g and the crop camera k, we cast one ray through each of 16×16 sub-patch centres of the crop and place 32 stratified samples per ray within $\pm r_o$ of the object centre's depth. Standard volume rendering [28] gives per ray the weights of the samples, the opacity o (the sum of the weights) and the expected latent of the surface the ray hits, $\bar{\mathbf{h}} = \sum_j w_j \mathbf{h}(x_j)/o$. A head maps $\bar{\mathbf{h}}$ and the view direction in the object frame to a 384-dimensional token feature; rays that miss the object see a background feature. Both are modulated by an appearance code a computed from the *observation's own* token statistics (per-channel mean and standard deviation over its 64 tokens), in the spirit of per-image appearance codes [27]. The code absorbs global lighting and background, and it is the same for every pose hypothesis of an observation, so it cannot by itself rank poses. Feature, opacity and latent are average-pooled 2×2 to the 8×8 token grid:

$$\big(\hat{\mathbf{z}}(g),\, o(g),\, \mathbf{h}(g)\big) = \mathcal{F}_\phi(g,\, k,\, a) \in \mathbb{R}^{64\times384}\times\mathbb{R}^{64}\times\mathbb{R}^{64\times128}. \tag{1}$$

The rendering is exact in the pose: a rotation of the hypothesis by 1° moves every ray's samples by 1° in the object frame. Nothing in $\hat{\mathbf{z}}$ comes from another image.

Box size. The field needs only a bound on the object's extent, no surface. On SPEED+ the box is fixed at $r_o = 1\,\mathrm{m}$ around the spacecraft origin. On T-LESS, r_o is 1.2 times the object radius listed in the dataset's model information; this single scalar per object is the only quantity taken from the CAD models (Sec. G).

Energy. The field defines an energy of a pose hypothesis given the observation, the foreground-weighted squared token distance

$$E(g; \mathbf{z}) = \frac{\sum_p w_p \, \frac{1}{D} \|\hat{\mathbf{z}}_p(g) - \mathbf{z}_p\|^2}{\sum_p w_p}, \tag{2}$$

with $D = 384$.

4.3. Fitting the world model

The field is fitted to the cached tokens of the training crops at their ground-truth poses, by minimising $E(g^\star; \mathbf{z})$ of Eq. (2) averaged over the training set, with batch 32, Adam and a cosine learning-rate schedule (60k steps on SPEED+, 100k on T-LESS). The loss is reconstruction only: there is no contrastive or ranking term, and the field never sees a wrong pose during fitting. On SPEED+ all token weights are 1; on T-LESS the weights w_p emphasise the visible object. The density, and with it the object's shape, is never supervised; it is whatever best explains the tokens across all training poses. The field is fitted once per dataset and is not updated afterwards.

4.4. Comparing inside every denoising step

At step t, the field renders Eq. (1) at each sample's pose g_t in its observation's crop. The denoiser forms one input token per image patch from the observation, the rendering, their difference, the weight and the field's latent,

$$x_p = \mathrm{Lin}\big([\,\mathrm{LN}(\mathbf{z}_p),\, \mathrm{LN}(\hat{\mathbf{z}}_p),\, \mathrm{LN}(\hat{\mathbf{z}}_p - \mathbf{z}_p),\, w_p,\, \mathrm{LN}(\mathbf{h}_p)\,]\big) + e_p, \tag{3}$$

with a learned position embedding e_p and all rendered quantities at g_t. The latent $\mathbf{h}_p(g_t)$ tells the denoiser which part of the object, in the field's own coordinates, the hypothesis places under patch p. It depends on the hypothesis only, not on the observation, and it is never decoded into coordinates or depth. A conditioning vector $c = \mathrm{Lin}(\mathrm{SiLU}(\gamma_t(t) + \gamma_g(r_t)[+\gamma_k(k)]))$ combines Fourier embeddings γ of the step and of the Lie-algebra coordinates r_t of g_t. On SPEED+ it also includes the crop camera; on T-LESS, as in LiePose, the crop camera enters only through the rendering. c is prepended as a token and modulates four adaLN [37] transformer blocks of width 256. The output at the conditioning token is mapped to the 6-dimensional noise ϵ_θ. The rendering is treated as a constant: no gradient reaches $\mathcal{F}_\phi$.

The network is trained and sampled with the generative process of Sec. 3 (noise-prediction loss, 100-step sampler,

weight averaging). Because the rendering depends on g_t, the denoiser can read the error of its own hypothesis from the difference tokens instead of inferring it from image features alone. The rendering costs one field evaluation per ray sample, sample and step; we report the cost in Sec. 5.5.

Control without the world model. To isolate the comparison, the control is the same network with input tokens $[\mathrm{LN}(\mathbf{z}_p),\ w_p]$, trained identically. It sees the same DINOv2 observation tokens, conditioning and pose embedding, so any difference between LaRC and the control is due to the rendered quantities alone. The control is a DINOv2 token denoiser, not LiePose, which uses a trained ResNet-34 encoder.

Trained encoder. The comparison is defined in frozen DINOv2 space, because that is the space the field renders. The denoiser can still use a trained image encoder for its own features. In this variant a frozen DINOv2 copy gives $\mathbf{z}$ for the comparison, and a second copy, trained with a $10\times$ lower learning rate, provides its 16×16 tokens; each 2×2 cell is concatenated, projected and added to x_p. The control with a trained encoder is built the same way. These rows are marked ‡.

4.5. Selecting among samples

A diffusion sampler returns one pose per run, and independent runs can land in different modes, *e.g.* flipped configurations of a near-symmetric object. We draw N samples from independent priors and return $\hat{g} = \arg\min_i E(g_i; \mathbf{z})$ (Eq. (2)). Selection uses only the observation and the field. We also report a selection rule that uses no world model, *consensus*: the sample with the most other samples within $10°$ in rotation. $N = 16$ for all benchmarks; the main tables use one sample unless marked otherwise.

4.6. Information used at test time

At test time LaRC uses the image crop, its crop camera k (box and intrinsics), the object identity, which selects the object's tri-planes as in the BOP task where the target objects are given, and the frozen field. No training image is needed at test time. Crops are ground-truth boxes on both benchmarks. On T-LESS these are the visible box and visible mask, as in LiePose's protocol [14]; on SPEED+ the box of the spacecraft keypoints projected with the ground-truth pose. Every method we train uses the same crops. Rows whose crops come from a detector are marked. Apart from these crops, no ground-truth quantity enters inference.

5. Experiments

5.1. Setup

Benchmarks. *T-LESS* [10]: 30 texture-less industrial objects with discrete and continuous symmetries, heavy clutter and occlusion. We train on the PBR and real training images and test on the BOP19 targets of the real test set. *YCB-V* [47]: 21 household objects in cluttered video frames; PBR and real training images, BOP19 test targets. **[TODO: YCB-V: fit the field and train; until then its columns stay empty.]** *SPEED+* [35]: a single spacecraft. We train only on the synthetic images and test on the full hardware-in-the-loop *lightbox* and *sunlamp* sets, without using their images in any form during training.

Metrics. On T-LESS and YCB-V we report the BOP average recalls of VSD, MSSD and MSPD and their mean AR [13], computed with the BOP toolkit; VSD uses the toolkit's Python renderer (Sec. E). On SPEED+ we report the mean translation error e_t (m), the mean rotation error $e_q = 2\arccos|\langle q, \hat{q}\rangle|$ (deg), and the official SPEC2021 score $S = \frac{1}{n}\sum(e_q^{\mathrm{rad}} + e_t/\|t\|)$ with the competition's precision thresholds [35] (Sec. E), on the full test sets.

Baselines and training budget. The main baseline is LiePose [14], which we retrain with its public code. The matched budget is LiePose's number of training images: 400k steps with batch 64. Because every noisy pose of LaRC queries the world model, we use 64 noisy poses per image instead of LiePose's 256; the control uses the same 64. Ablations use 100k steps with batch 32 and 16 noisy poses per image for every row, an eighth of the matched number of training images. Our T-LESS LiePose run uses batch 64 where the paper uses 32, and the control and LaRC draw from a fixed set of pre-augmented crops (Sec. G). We evaluate the final checkpoint of every run; no checkpoint is chosen on test data. Published numbers of other methods are copied from their papers and marked by protocol.

Implementation. The field has three $128 \times 128 \times 32$ tri-planes per object, an MLP of width 128 and 32 samples per ray, and is fitted for 60k (SPEED+) or 100k (T-LESS) steps. The denoiser has 4 blocks of width 256. Sampling uses LiePose's 100-step sampler. Full details are in Sec. A.

5.2. Main results

T-LESS and YCB-V. Table 1 compares LaRC with LiePose trained by us under identical crops, data and training budget, with the no-world-model control, and with published methods. **[TODO: describe once the rows are filled: (i) LaRC vs control (the effect of the comparison), (ii) LaRC vs LiePose (ours), (iii) where the difference**

Table 1. **T-LESS and YCB-V** (BOP19 targets, BOP average recall, %). *Crops*: GT = ground-truth visible box and mask; Det = detector boxes. Rows with different crops are not directly comparable. *Ours* = trained by us with the same data, crops and number of training images. Published numbers are rounded to one decimal. †AR not reported; mean of the three recalls, computed by us. The GDRNPP numbers that [14] compares with are GDRNPP's leaderboard entry with one model per dataset and detected boxes (67.6 / 75.1 / 90.2), not a ground-truth-crop result. Published rows train on PBR and real images; §per-metric recalls from the BOP leaderboard [1], as the papers report AR only (GDRNPP) or no BOP numbers (ZebraPose). ‡Encoder trained (a frozen copy gives the comparison space). Results with the retrieval-referenced world model are in Tab. 7. **[TODO: YCB-V of our rows.]**

				T-LESS				YCB-V			
Method	Encoder	Crops	Samples	VSD	MSSD	MSPD	AR	VSD	MSSD	MSPD	AR
Published, detector crops (BOP leaderboard, RGB only)											
CosyPose [19]	–	Det	1	66.9	69.5	82.1	72.8	77.2	84.2	85.0	82.1
SurfEmb [9]	–	Det	1	–	–	–	77.0	–	–	–	71.8
ZebraPose [40]§	–	Det	1	69.6	74.0	88.9	77.5	75.1	83.0	86.4	81.5
CIR, RGB [25]	–	Det	1	66.3	68.4	79.8	71.5	78.3	83.5	85.2	82.4
GDRNPP [26]§	–	Det	1	68.7	76.0	91.3	78.6	76.0	84.6	86.9	82.5
Published, ground-truth crops											
LiePose [14]	ResNet-34	GT	1	56.9	60.2	93.2	70.1†	–	–	–	–
Trained by us, ground-truth crops, matched number of training images (400k × 64)											
LiePose (our training)	ResNet-34	GT	1	63.1	65.7	92.3	73.7				
Control (no world model)	DINOv2-S	GT	1								
LaRC	DINOv2-S	GT	1								
LaRC + selection	DINOv2-S	GT	*N*								
Trained by us, ground-truth crops, 100k × 32 (an eighth of the training images, 16 noisy poses)											
Control (no world model)	DINOv2-S	GT	1	21.5	24.9	75.2	40.5				
LaRC	DINOv2-S	GT	1	65.3	65.5	86.5	72.4				
Control, trained encoder	DINOv2-S‡	GT	1								
LaRC, trained encoder	DINOv2-S‡	GT	1								

comes from, using the rotation and translation accuracies of Tab. 3, (iv) the effect of selection. Do not compare GT-crop rows with detector rows.] At 100k × 32, LaRC raises AR on T-LESS from 40.5 to 72.4 over the control, with all three recalls higher (Tab. 1). The gain is largest in translation, where the recall at 2 cm rises from 26.8 to 80.3, and the rotation recall at 5° rises from 45.7 to 71.6 (Tab. 3). With an eighth of the training images, LaRC is 1.3 AR below LiePose trained by us at the full budget (73.7); this is one seed, and the matched-budget rows are not yet run. At the same budget the retrieval world model of Sec. I raises AR from 41.2 to 54.7 over its own control (Tab. 7).

SPEED+. Table 2 reports the full lightbox and sunlamp sets. At 100k × 32, LaRC lowers the score from 0.827 to 0.705 on lightbox and from 0.743 to 0.643 on sunlamp relative to the control, with lower rotation and translation errors on both. Both are far from LiePose at the full budget (0.234 / 0.289). With a trained encoder (Sec. 4.4) and 100k steps of batch 64, LaRC reaches 0.412 / 0.480. This run differs from the rows above in both the encoder and the batch, and its control with a trained encoder has not been evaluated, so the two effects are not separated. **[TODO: describe the matched-budget rows, the trained encoder and selection, and the gap to the oracle as an upper bound on what selection could reach. State explicitly that the SPEC2021 entries used the unlabeled test-domain images and are not under our protocol.]**

5.3. Ablations

What the denoiser reads from the field. Table 3 varies the denoiser's input tokens with the field fixed. Adding the rendered tokens and their difference to the observation lowers the SPEED+ score from 0.827 to 0.742 (lightbox) and from 0.743 to 0.687 (sunlamp); adding the field's latent lowers it further, to 0.705 and 0.643, which is LaRC. Two further rows test whether the comparison alone suffices. With only per-token comparison statistics and no observation features, the denoiser is far worse than the control (1.516 / 1.507). Adding the latent recovers part of this (1.318 / 1.326). Training with the raw channels dropped for half the images, so that the denoiser must also pose from the statistics, does not help (0.758 / 0.724). The observation's own features are therefore needed: the field's rendering is not precise enough to replace them, and the denoiser uses it as an additional input. **[TODO: T-LESS column; energy-only and oracle rows.]**

Table 2. **SPEED+** (full lightbox and sunlamp sets; e_t in m, e_q in degrees, S = official SPEC2021 score; lower is better). All methods train on synthetic images only, except where marked. *Box*: GT = box of the keypoints projected with the ground-truth pose; Det = detected. The oracle row selects with the ground truth and is a diagnostic, not a result. S applies the precision floors to rotation and translation separately, as the competition server does; *floors applied jointly as in [35] (at most 0.005 lower per image). –: not reported in metres. Published e_q rounded to 0.1°. EagerNet uses test-time augmentation and several hypotheses. ¶Best submission of each team per category on the final leaderboard [2], e_q as tabulated in [34]. Consensus: of 16 samples, the one with the most others within 10° (no ground truth, no world model). ‡Encoder trained (a frozen copy gives the comparison space). Results with the retrieval-referenced world model are in Tab. 8.

			lightbox			sunlamp		
Method	Box	Samples	e_t	e_q	S	e_t	e_q	S
Published, synthetic training only								
KRN [35]*	GT	1	2.25	44.5	1.12	14.64	81.0	3.73
HigherHRNet + EPnP [35]*	–	1	0.97	34.7	0.77	0.85	47.8	0.98
SPNv2 (ϕ=3, BN) [33]	–	1	0.175	6.5	0.142	0.225	11.1	0.230
EagerNet [43]	Det	–	–	1.7	0.039	–	2.7	0.059
SPNv3-B [34]	GT	1	–	2.2	0.051	–	3.6	0.079
Legrand et al. (HRNet) [22]	GT	1	0.09	4.3	0.09	0.14	6.9	0.14
Published, using unlabeled lightbox/sunlamp images (different protocol)								
SPNv2 + online domain refinement [33]	–	1	0.142	5.6	0.122	0.182	9.6	0.198
SPEC2021, TangoUnchained [36]¶	–	1	–	3.2	0.073	–	4.3	0.090
SPEC2021, lava1302 [36]¶	–	1	–	6.7	0.165	–	2.7	0.059
Trained by us, synthetic only, GT keypoint box, matched number of training images (400k × 64)								
LiePose (our training)	GT	1	0.183	11.6	0.234	0.175	14.9	0.289
+ consensus	GT	16	0.170	7.8	0.165	0.163	11.2	0.223
Control (no world model)	GT	1						
LaRC	GT	1						
LaRC + selection	GT	N						
+ consensus	GT	16						
Trained by us, 100k × 32 (an eighth of the training images, 16 noisy poses)								
Control (no world model)	GT	1	0.358	43.8	0.827	0.308	39.6	0.743
LaRC	GT	1	0.285	37.5	0.705	0.244	34.5	0.643
Trained by us, trained encoder‡, 100k × 64 (a quarter of the steps, 64 noisy poses)								
Control (no world model)	GT	1						
LaRC	GT	1	0.155	22.1	0.412	0.219	25.4	0.480
Oracle selection among the same N (diagnostic)	GT	N						

Selection. Table 4 varies the number of samples. The oracle column shows how often a good sample exists among the N, and therefore how much any selection rule could gain; the energy column shows how much of this the field recovers. **[TODO: describe.]**

5.4. Analysis

What the world model can and cannot see. Table 5 measures the world model on its own, independently of the denoiser: how reliably its energy prefers the true pose over perturbed ones. On SPEED+ the field prefers the true pose over perturbations of 3–10° in 84% of cases and of 10–30° in 98%, and over its 180° twins in 98%. Below 1° it is at chance (0.52). **[TODO: retrieval on SPEED+ under the same protocol.]** On T-LESS the field is no better than retrieval at fine rotations (0.53 at 1–3°), slightly better at 3–10° and in translation, and worse at 10–30° (0.86 vs 0.92). The pose its energy picks among candidates has a median error of 3.4° but a mean of 14.2°, against a mean of 7.2° for retrieval (Tab. 6): the field makes more gross errors. We have not established why the field is weaker on T-LESS than on SPEED+; the objects are small, texture-less and many, while SPEED+ has one large, textured object. **[TODO: relate the finest bin to the remaining error of LaRC in Tabs. 1 and 2.]**

When are flips decided? On SPEED+, the dominant failure is a flipped estimate. Figure 3 tracks, along the sampling trajectory, the share of samples that are flipped and how well the energy separates flipped from correct samples of the same image. **[TODO: describe; this analysis motivates selection (Sec. 4.5).]**

Table 3. **What the denoiser reads from the field.** Same denoiser and budget (100k steps × batch 32, 16 noisy poses per image), one sample, frozen DINOv2. T-LESS: AR (%, ↑); SPEED+: score S on lightbox / sunlamp (↓). Statistics: per token, the cosine and squared distance between $\hat{\mathbf{z}}_p$ and $\mathbf{z}_p$, the opacity o_p, and the cosines of $\mathbf{z}_p$ with the 9 rendered tokens around p. **[TODO: fill the T-LESS column.]**

Denoiser input per token	T-LESS	SPEED+
$[\mathbf{z}, w]$ (control)	40.5	0.827 / 0.743
$[\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}, w]$		0.742 / 0.687
$[\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}, w, \mathbf{h}]$ (LaRC)	72.4	0.705 / 0.643
+ statistics, raw channels dropped (p=0.5)		0.758 / 0.724
Statistics only (no $\mathbf{z}$)		1.516 / 1.507
Statistics + $\mathbf{h}$ (no $\mathbf{z}$)		1.318 / 1.326
Energy only (control + selection)		
Oracle $\hat{\mathbf{z}}$: GT-view tokens (diag.)		

Table 4. **Selection among N samples** (SPEED+ lightbox S, ↓; T-LESS AR, ↑), LaRC at the matched budget. Energy: Eq. (2); consensus: Sec. 4.5; oracle: the sample closest to the ground truth in rotation (diagnostic). Selection with the retrieval world model is in Tab. 12. **[TODO: fill.]**

	SPEED+ lightbox S			T-LESS AR		
N	consensus	energy	oracle	consensus	energy	oracle
1						
4						
16						

Table 5. **What the world model can discriminate.** For held-out observations, the probability that the energy of a perturbed pose exceeds that of the ground-truth pose (0.5 = chance), by size of the rotation perturbation, and for translation perturbations. Grey: the retrieval-referenced world model (Sec. I), references retrieved at a noisy ground truth. T-LESS: the same 500 held-out training crops for both, which also chose each model's checkpoint. SPEED+: 300 crops of the synthetic validation split, 6 perturbations per bin; this split also chose the field's checkpoint.

	rotation perturbation			
	1–3°	3–10°	10–30°	translation
T-LESS, field	0.53	0.66	0.86	0.66
T-LESS, retrieval	0.53	0.64	0.92	0.60
SPEED+, field	0.65	0.84	0.98	
SPEED+, retrieval				
YCB-V, field				

5.5. Runtime

Table 6 breaks down the inference cost. The field renders 16 × 16 rays with 32 samples once per sample and step. **[TODO: describe; mention fewer steps as in LiePose's runtime study if measured.]**

FIGURE PLACEHOLDER (flip timing, SPEED+ lightbox, diagnostic).
x: denoising step (0 = pure noise, 100 = final). Left y: share of samples whose rotation error exceeds 90°. Right y: within-image probability that the energy prefers a good sample (< 30°) over a flipped one (> 90°). Curves for LaRC and control.

Figure 3. **When are flips decided?** The share of flipped samples and the field's ability to tell flipped from correct hypotheses, along the sampling trajectory. Ground truth is used only to measure. **[TODO: caption text from the result.]**

FIGURE PLACEHOLDER (qualitative).
Rows: T-LESS, SPEED+ lightbox, SPEED+ sunlamp. Columns: input crop; LiePose estimate (projected model outline); LaRC estimate; rendered tokens at the estimate vs observed tokens (PCA colours); rendered opacity; one failure case per dataset.

Figure 4. **Qualitative results and failures. [TODO: caption.]**

Table 6. **Runtime** per image (ms), one sample, 100 steps, batch 100, on one **[TODO: GPU model]**. **[TODO: fill.]**

Method	encoder	field	denoiser	total
LiePose (ours)		–		
Control		–		
LaRC				
LaRC + selection (N)				

6. Limitations and Conclusion

Limitations. LaRC relies on ground-truth crops on both benchmarks, as LiePose does on T-LESS; with detected boxes and masks the inputs of the field and of the denoiser change, and we have not evaluated this setting. The field is fitted per object from posed training crops, needs the object's identity and the crop camera at test time, and does not apply to unseen objects. On T-LESS it uses one scalar per object from the dataset's model information, the box size. Its precision is bounded by what frozen, pooled DINOv2 tokens can distinguish and by how well a field fitted from tokens alone captures each object (Tab. 5): as a ranker on its own, on T-LESS it makes more gross errors than a retrieval-referenced world model. Rendering at every step makes sampling slower than LiePose (Tab. 6). Finally, our runs use

64 noisy poses per training image instead of LiePose's 256. **[TODO: revise once the results are in; add any failure mode seen in Fig. 4.]**

Conclusion. We moved render-and-compare into SE(3) diffusion. A feature field of the object, fitted once to frozen DINOv2 tokens, renders at every denoising step and for every sample the tokens of the object at the current hypothesis, and the denoiser conditions on their comparison with the observation and on the field's latent. The same rendering gives an energy that ranks samples without ground truth. Because the diffusion formulation is unchanged and the control differs only by the rendered quantities, the experiments isolate what the comparison contributes. **[TODO: one or two sentences of findings, written after the tables are filled.]**

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